Diagonalization of a Matrix

August 31, 2025

Similar Matrices and Diagonalization

- **Similar Matrices:** Square matrices A and B are similar if there exists an invertible matrix P such that $A = PBP^{-1}$.
- **Diagonalizable Matrix:** A square matrix A is diagonalizable if it is similar to a diagonal matrix.

Characteristic Polynomial of Similar Matrices

Theorem

If A and B are similar matrices, then they share the same characteristic polynomial.

Condition for Diagonalization

Theorem

A square matrix A of order n is diagonalizable if and only if it has n linearly independent eigenvectors.

Note on Distinct Eigenvalues

- If a square matrix A of order n has n distinct eigenvalues, then it is diagonalizable.

Working Procedure: Diagonalization of a Matrix A

1. **Find all eigenvalues of A:** Solve the characteristic equation $\det(A - \lambda I) = 0$.
2. **Find n linearly independent eigenvectors of A:** For each eigenvalue λ, find a basis for the eigenspace $E_\lambda(A)$ by solving $(A - \lambda I)X = 0$.
3. **Construct P and D:** If $v_1, v_2, \ldots, v_n$ are linearly independent eigenvectors corresponding to eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, respectively, then set the matrix $P = [v_1|v_2|\ldots|v_n]$ and $D = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$.
4. **Verify:** Then $A = PDP^{-1}$.

Problem 1: Diagonalize and find $\sqrt{A}$

Problem: Diagonalize the matrix $A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$. Hence find $\sqrt{A}$.

Problem 1: Solution

- **Characteristic polynomial:** $\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 1 \\ 2 & 2-\lambda \end{pmatrix} =$ $(1-\lambda)(2-\lambda) - 2 = 2 - 3\lambda + \lambda^2 - 2 = \lambda^2 - 3\lambda = \lambda(\lambda - 3)$.
- **Eigenvalues:** $\lambda_1 = 0$, $\lambda_2 = 3$.
- **Eigenvector for $\lambda_1 = 0$:** Solve $(A - 0I)X = 0 \Rightarrow \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. This gives $x + y = 0$. So, $y = -x$. A basis vector is $v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.
- **Eigenvector for $\lambda_2 = 3$:** Solve $(A - 3I)X = 0 \Rightarrow \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. This gives $-2x + y = 0$. So, $y = 2x$. A basis vector is $v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.
- **Construct P and D:** $P = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}$ and $D = \begin{pmatrix} 0 & 0 \\ 0 & 3 \end{pmatrix}$.

Problem 1: Solution (Contd.)

- **Find P^{-1}:**
$$P^{-1} = \frac{1}{1(2)-1(-1)} \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2/3 & -1/3 \\ 1/3 & 1/3 \end{pmatrix}.$$
- **Diagonalization:** $A = PDP^{-1}$.
- **To find $\sqrt{A}$:** $\sqrt{A} = P\sqrt{D}P^{-1}$. Here
$$\sqrt{D} = \begin{pmatrix} \sqrt{0} & 0 \\ 0 & \sqrt{3} \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{3} \end{pmatrix}.$$

$$\begin{aligned} \sqrt{A} &= \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{3} \end{pmatrix} \begin{pmatrix} 2/3 & -1/3 \\ 1/3 & 1/3 \end{pmatrix} \\ &= \begin{pmatrix} 0 & \sqrt{3} \\ 0 & 2\sqrt{3} \end{pmatrix} \begin{pmatrix} 2/3 & -1/3 \\ 1/3 & 1/3 \end{pmatrix} \\ &= \begin{pmatrix} \sqrt{3}/3 & \sqrt{3}/3 \\ 2\sqrt{3}/3 & 2\sqrt{3}/3 \end{pmatrix} \end{aligned}$$

Problem 2 : Determine if the matrix is diagonalizable

Problem: Determine if the matrix is diagonalizable : $A = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{pmatrix}$.

Problem 2 : Solution

- After calculating the determinant of $(A - \lambda I)$, we get the equation:

$$-\lambda^3 + \lambda^2 = 0 \implies \lambda^2(1 - \lambda) = 0$$

- The eigenvalues are $\lambda_1 = 0$(multiplicity 2) and $\lambda_2 = 1$ (multiplicity 1)

Problem 2 : Solution – Find Eigenvectors for $\lambda_1 = 0$

- We look for vectors $\mathbf{v}$ such that $A\mathbf{v} = 0\mathbf{v}$, or $(A - 0I)\mathbf{v} = \mathbf{0}$.
- Solving the system:

$$\begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- We find that any such vector must be of the form $\begin{bmatrix} z \\ z \\ z \end{bmatrix}$, which means it's a multiple of $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.
- There is only one independent vector in this case.

Problem 2 : Solution – Find the Eigenvectors for $\lambda_2 = 1$

- We look for vectors $\mathbf{v}$ such that $A\mathbf{v} = 1\mathbf{v}$, or $(A - 1I)\mathbf{v} = \mathbf{0}$.
- Solving the system:

$$\begin{bmatrix} -1 & -1 & 1 \\ -1 & -1 & 1 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- We find that any such vector must be of the form $\begin{bmatrix} 0 \\ z \\ z \end{bmatrix}$, which means it's a multiple of $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.
- There is also only **one independent vector** in this case.

Problem 2 : Solution – Conclusion

- We need three independent eigenvectors for a 3x3 matrix to be diagonalizable. Since we only found two, the matrix A is not diagonalisable.

Problem 3: Diagonalize and find A^{100}

Problem: Diagonalize the matrix $A = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$. Hence find A^{100}.

Problem 3: Solution

- **Characteristic polynomial:** The characteristic polynomial is $\Delta(A) = \lambda^3 - \lambda$.
- **Eigenvalues:** $\lambda_1 = -1$, $\lambda_2 = 0$, $\lambda_3 = 1$.

Problem 3: Solution(contd.)

- **Eigenspace for $\lambda_1 = -1$:** Solve $(A - (-1)I)X = 0 \Rightarrow (A + I)X = 0$.
$\begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. Row operations lead to
$x - y + z = 0$, $-x + y + z = 0$ and $x - y + z = 0$. So $x = y$ and $z = 0$. $E_{-1}(A) = \{(x, x, 0) : x \in \mathbb{R}\}$. A basis is $v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.
- **Eigenspace for $\lambda_2 = 0$:** Solve $(A - 0I)X = 0 \Rightarrow AX = 0$.
$\begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. This gives $x = y = z$.
$E_0(A) = \{(x, x, x) : x \in \mathbb{R}\}$. A basis is $v_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

Problem 3: Solution

- **Eigenspace for $\lambda_3 = 1$:** Solve $(A - 1I)X = 0 \Rightarrow (A - I)X = 0$.
$\begin{pmatrix} -1 & -1 & 1 \\ -1 & -1 & 1 \\ 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. This gives $y = 0$ and $x = z$.

  $E_1(A) = \{(x, 0, x) : x \in \mathbb{R}\}$. A basis is $v_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$.

- **Construct P and D:** $P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ and $D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

- **Find P^{-1}:** $P^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 0 \end{pmatrix}$.

Problem 3: Solution

- **Diagonalization:** $A = PDP^{-1}$.
- **To find A^{100}:** $A^{100} = PD^{100}P^{-1}$.

$$D^{100} = \begin{pmatrix} (-1)^{100} & 0 & 0 \\ 0 & 0^{100} & 0 \\ 0 & 0 & 1^{100} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$A^{100} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}$$

Problem 4: Diagonalize a matrix

Problem: Diagonalize the matrix $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix}$.

Problem 4 : Solution

- **Characteristic polynomial:**
  $\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 1 & 1 \\ 2 & 2-\lambda & 2 \\ 1 & 1 & 1-\lambda \end{pmatrix}$. Expanding this, we get
  $\lambda^3 - 4\lambda^2 = 0 \Rightarrow \lambda^2(\lambda - 4) = 0$.
- **Eigenvalues:** $\lambda_1 = 0$ (multiplicity 2), $\lambda_2 = 4$ (multiplicity 1).

Problem 4: Solution

Eigenspace for $\lambda_1 = 0$: Solve $AX = 0$. $\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. This gives $x + y + z = 0$. The eigenspace $E_0(A)$ has dimension 2. We need two linearly independent vectors satisfying $x + y + z = 0$. We can choose $(y, z) = (1, 0) \Rightarrow x = -1$. So $v_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$. We can choose $(y, z) = (0, 1) \Rightarrow x = -1$. So $v_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$. These two vectors are linearly independent.

Problem 4: Solution:

- **Eigenspace for $\lambda_2 = 4$:** Solve $(A - 4I)X = 0$.
$\begin{pmatrix} -3 & 1 & 1 \\ 2 & -2 & 2 \\ 1 & 1 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. Using Gauss elimination:
$R_2 \leftarrow R_2 + \frac{2}{3}R_1$: $\begin{pmatrix} -3 & 1 & 1 \\ 0 & -4/3 & 8/3 \\ 1 & 1 & -3 \end{pmatrix}$ $R_3 \leftarrow R_3 + \frac{1}{3}R_1$:
$\begin{pmatrix} -3 & 1 & 1 \\ 0 & -4/3 & 8/3 \\ 0 & 4/3 & -8/3 \end{pmatrix}$ $R_3 \leftarrow R_3 + R_2$: $\begin{pmatrix} -3 & 1 & 1 \\ 0 & -4/3 & 8/3 \\ 0 & 0 & 0 \end{pmatrix}$ From the second row: $-\frac{4}{3}y + \frac{8}{3}z = 0 \Rightarrow -y + 2z = 0 \Rightarrow y = 2z$. From the first row:
$-3x + y + z = 0 \Rightarrow -3x + 2z + z = 0 \Rightarrow -3x + 3z = 0 \Rightarrow x = z$. So $E_4(A) = \{(z, 2z, z) : z \in \mathbb{R}\}$. A basis is $v_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$.

Problem 4: Solution

- **Construct P and D:** $P = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix}$ and $D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{pmatrix}$.

  Since A has 3 linearly independent eigenvectors, it is diagonalizable.

  Then $P^{-1} = \begin{pmatrix} -1/2 & 1/2 & -1/2 \\ -1/4 & -1/4 & 3/4 \\ 1/4 & 1/4 & 1/4 \end{pmatrix}$ Hence, $A = PDP^{-1}$

Orthogonally Diagonalizable Matrices

- A square matrix A is **orthogonally diagonalizable** if there exists an orthogonal matrix P such that $A = PDP^T$, where D is a diagonal matrix.
- An orthogonal matrix P is a square matrix whose columns (and rows) are orthogonal unit vectors, meaning $P^T P = PP^T = I$, so $P^{-1} = P^T$.

Spectral Theorem for Symmetric Matrices

Theorem

Let A be a symmetric matrix of order n. Then:

1. *A has n real eigenvalues, counting multiplicities.*
2. *If λ is an eigenvalue of A with multiplicity k, then the dimension of the eigenspace $E_\lambda(A)$ is k. (The geometric multiplicity equals the algebraic multiplicity).*
3. *The eigenvectors corresponding to different eigenvalues are orthogonal.*
4. *A is orthogonally diagonalizable.*

Problem 5: Orthogonal Diagonalization

Problem: Find an orthogonal matrix that diagonalizes the matrix

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Problem 5 : Solution

- **Characteristic polynomial:**
  $\det(A-\lambda I) = \det\begin{pmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{pmatrix} = -\lambda(\lambda^2-1) - 1(-\lambda-1) +$
  $1(1+\lambda) = -\lambda^3+\lambda+\lambda+1+1+\lambda = -\lambda^3+3\lambda+2 = -(\lambda^3-3\lambda-2).$
  Factoring gives $-(\lambda+1)^2(\lambda-2)=0.$
- **Eigenvalues:** $\lambda_1 = -1$ (multiplicity 2), $\lambda_2 = 2$ (multiplicity 1).

Problem 5 :Solution

- **Eigenspace for $\lambda_1 = -1$:** Solve $(A - (-1)I)X = 0 \Rightarrow (A + I)X = 0$. $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. This gives $x + y + z = 0$. The eigenspace $E_{-1}(A)$ has dimension 2. Two linearly independent vectors satisfying $x + y + z = 0$ are $v_1' = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ and $v_2' = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$. These are not orthogonal ($v_1'^T v_2' = 1 \neq 0$). Apply Gram-Schmidt: $u_1 = \frac{v_1'}{\|v_1'\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$. $w_2 = v_2' - (v_2'^T u_1)u_1 = \begin{pmatrix} 1/2 \\ 1/2 \\ -1 \end{pmatrix}$. To avoid fractions, lets assume $w_2 = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$

Problem 5: Solution

- $u_2 = \frac{w_2}{||w_2||} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$.
- **Eigenspace for $\lambda_2 = 2$:** Solve $(A - 2I)X = 0$.
  $\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. Row operations lead to $-2x + y + z = 0$, $x - 2y + z = 0$ and $x + y - 2z = 0$. So $y = z$ and $x = y$. $E_2(A) = \{(x, x, x) : x \in \mathbb{R}\}$. A basis is $v_3' = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. Normalize v_3': $u_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

Problem 5 : Solution

- **Orthogonal matrix P:**
  $P = [u_1|u_2|u_3] = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 0 & -2/\sqrt{6} & 1/\sqrt{3} \end{pmatrix}$. This P orthogonally diagonalizes A.

Problem 6: Orthogonal Diagonalization

Problem: Find an orthogonal matrix that diagonalizes the matrix $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$.

Problem 6: Solution

- **Characteristic polynomial:** $\det(A - \lambda I) = \det \begin{pmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 4 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3)$.
- **Eigenvalues:** $\lambda_1 = 1$, $\lambda_2 = 3$.
- **Eigenspace for $\lambda_1 = 1$:** Solve $(A - 1I)X = 0$.
  $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. This gives $x - y = 0 \Rightarrow x = y$.
  $E_1(A) = \{(x, x) : x \in \mathbb{R}\}$. A basis is $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Normalize:
  $u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Problem 6: Solution

- **Eigenspace for $\lambda_2 = 3$:** Solve $(A - 3I)X = 0$.
  $\begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. This gives $-x - y = 0 \Rightarrow y = -x$.
  $E_3(A) = \{(x, -x) : x \in \mathbb{R}\}$. A basis is $v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Normalize:
  $u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.
- **Orthogonal matrix P:** $P = [u_1|u_2] = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$. This P orthogonally diagonalizes A.